Which OSI layer is responsible for logical addressing and routing?
A. Physical layer
B. Session layer
C. Presentation layer
D. Network layer
ANSWER: D

Which OSI layer provides end-to-end transport services?
A. Physical layer
B. Presentation layer
C. Data Link layer
D. Transport layer
ANSWER: D

Which OSI layer is most closely associated with Ethernet frame forwarding on a LAN?
A. Presentation layer
B. Application layer
C. Data Link layer
D. Session layer
ANSWER: C

What is the main purpose of internetworking?
A. Converting every network into a wireless LAN
B. Connecting separate networks so devices can communicate
C. Preventing routing between networks
D. Replacing all IP addresses with MAC addresses
ANSWER: B

Which Ethernet address identifies a network interface at Layer 2?
A. IPv4 subnet mask
B. DNS alias
C. TCP port number
D. MAC address
ANSWER: D

What does a wireless access point commonly provide?
A. Wireless devices with access to a network
B. IP address translation in every design
C. Email mailbox storage
D. Disk partitioning
ANSWER: A

What is a common difference between UTP and STP cable?
A. UTP always uses fibre strands
B. STP includes shielding to reduce electromagnetic interference
C. UTP requires a router at each end
D. STP cannot carry Ethernet
ANSWER: B

What does the 'e' in Cat 5e indicate?
A. Extended Ethernet address
B. Enhanced Category 5 cabling
C. Encrypted Category 5
D. Electrical-only cable
ANSWER: B

Which connector is commonly used with twisted-pair Ethernet patch cables?
A. F-type coax connector
B. DB-9 serial connector
C. RJ-45 style 8P8C connector
D. SC fibre connector only
ANSWER: C

Why should cable categories be selected to match the network requirement?
A. Cable category determines the IP address
B. They differ in supported performance and transmission characteristics
C. Cable category replaces switch configuration
D. All categories have identical capabilities
ANSWER: B

Which OSI layer converts bits into electrical, optical, or radio signals?
A. Transport layer
B. Network layer
C. Physical layer
D. Application layer
ANSWER: C

Which OSI layer commonly uses switches to forward frames using MAC addresses?
A. Data Link layer
B. Application layer
C. Session layer
D. Presentation layer
ANSWER: A

What is the purpose of a router in an internetwork?
A. Assigning file permissions
B. Displaying web pages
C. Forwarding packets between IP networks
D. Regenerating only electrical signals
ANSWER: C

What is a collision domain in Ethernet switching?
A. An IP address pool
B. A wireless encryption key
C. A network segment where simultaneous transmissions can collide
D. A DNS zone
ANSWER: C

What is a broadcast domain?
A. The set of devices that receive a Layer 2 broadcast
B. All devices on the Internet
C. Only devices with the same MAC vendor
D. A single TCP connection
ANSWER: A

Which wireless security protocol is generally preferred over obsolete WEP?
A. WPA2 or WPA3, depending on device support
B. WEP
C. Telnet
D. Open authentication with no protection
ANSWER: A

What is a key advantage of twisted-pair cable pairs?
A. It assigns IP addresses
B. It replaces network switches
C. Twisting helps reduce electromagnetic interference and crosstalk
D. It eliminates all signal loss at any distance
ANSWER: C

What does STP shielding need to be considered alongside?
A. VLAN names only
B. TCP port numbers
C. DNS TTL
D. Proper grounding and compatible installation practices
ANSWER: D

What is a common purpose of an Ethernet switch?
A. Resolving public domain names
B. Forwarding frames within a LAN based on MAC learning
C. Encrypting every application
D. Leasing IP addresses by default
ANSWER: B

Why should wireless access points use appropriate channel planning?
A. To disable roaming
B. To replace authentication
C. To increase IPv4 address length
D. To reduce interference and improve wireless performance

ANSWER: D

What is the main role of IP?
A. Resolving host names only
B. Encrypting all application data
C. Addressing and forwarding packets between networks
D. Providing reliable byte-stream delivery by itself
ANSWER: C

Which TCP/IP layer includes TCP and UDP?
A. Physical layer only
B. Network access layer only
C. Transport layer
D. Application layer only
ANSWER: C

Which TCP feature helps provide reliable ordered delivery?
A. No connection state
B. Sequence numbers and acknowledgements
C. MAC address learning
D. Broadcast-only delivery
ANSWER: B

Which protocol is connectionless and has lower transport overhead?
A. HTTP
B. TLS
C. TCP
D. UDP
ANSWER: D

Which protocol is commonly used when reliable ordered delivery is needed?
A. TCP
B. UDP
C. ARP
D. ICMP only
ANSWER: A

Which TCP/IP layer includes HTTP, DNS, and SMTP?
A. Internet layer
B. Application layer
C. Network access layer
D. Transport layer
ANSWER: B

What is the purpose of an IP address?
A. To identify a TCP application process by itself
B. To identify a network interface for IP communication
C. To define a file permission
D. To encrypt an Ethernet frame
ANSWER: B

Which protocol is used to report network diagnostic and error information, such as echo requests?
A. SMTP
B. DHCP only
C. FTP
D. ICMP
ANSWER: D

What is encapsulation in networking?
A. Adding protocol headers as data moves down the network stack
B. Removing all headers before transmission
C. Assigning a username to a packet
D. Converting a switch into a router
ANSWER: A

What is a key difference between TCP and UDP?
A. TCP has no ordering mechanism
B. They are identical protocols
C. UDP always retransmits lost packets

D. TCP provides a connection-oriented reliable stream; UDP sends datagrams without built-in delivery guarantees
ANSWER: D

What does /24 mean in CIDR notation for IPv4?
A. 24 host bits
B. The address is IPv6
C. 24 octets
D. The first 24 bits are the network prefix
ANSWER: D

How many total IPv4 addresses are in a /24 subnet?
A. 128
B. 24
C. 512
D. 256
ANSWER: D

How many conventional usable host addresses are in a typical /24 IPv4 subnet?
A. 254
B. 255
C. 252
D. 256
ANSWER: A

What is the subnet mask for /30?
A. 255.255.255.252
B. 255.255.255.248
C. 255.255.255.0
D. 255.255.0.0
ANSWER: A

How many conventional usable host addresses are available in a /30 subnet?
A. 30
B. 4
C. 2
D. 6
ANSWER: C

What is VLSM used for?
A. Encrypting subnet masks
B. Replacing IP routing
C. Allocating subnets of different sizes according to host requirements
D. Making every subnet the same size
ANSWER: C

For 192.168.5.85/24, what is the network address?
A. 192.168.5.255
B. 192.168.5.0
C. 192.168.0.0
D. 192.168.5.85
ANSWER: B

For 10.128.240.50/30, what is the network address?
A. 10.128.240.52
B. 10.128.240.50
C. 10.128.240.0
D. 10.128.240.48
ANSWER: D

When planning VLSM, which subnet should generally be allocated first?
A. The subnet with the lowest device name
B. The subnet with the highest gateway number
C. The subnet requiring the fewest hosts
D. The subnet requiring the most host addresses
ANSWER: D

A subnet must support 52 usable IPv4 host addresses. Which prefix is the smallest typical fit?
A. /28

B. /29
C. /26
D. /27
ANSWER: C

What is the broadcast address for 192.168.5.0/24?
A. 192.168.5.0
B. 192.168.5.255
C. 192.168.5.1
D. 192.168.4.255
ANSWER: B

What is the usable host range for 192.168.5.0/24?
A. 192.168.5.0 through 192.168.5.255
B. 192.168.4.1 through 192.168.4.254
C. 192.168.5.1 through 192.168.5.255
D. 192.168.5.1 through 192.168.5.254
ANSWER: D

What is the subnet mask for /26?
A. 255.255.255.224
B. 255.255.255.0
C. 255.255.255.240
D. 255.255.255.192
ANSWER: D

How many usable host addresses does a typical /26 IPv4 subnet provide?
A. 62
B. 126
C. 30
D. 64
ANSWER: A

Which prefix provides 30 usable IPv4 host addresses in a typical subnet?
A. /27
B. /28
C. /26
D. /29
ANSWER: A

How many total addresses are in a /30 block?
A. 16
B. 4
C. 8
D. 2
ANSWER: B

Why is the broadcast address reserved in a conventional IPv4 subnet?
A. It is the DNS server address
B. It is always the router's address
C. It identifies the subnet mask
D. It represents delivery to all hosts on that subnet
ANSWER: D

What is the network address for 10.128.240.50/30?
A. 10.128.240.51
B. 10.128.240.48
C. 10.128.240.50
D. 10.128.240.52
ANSWER: B

What is the block size of a /28 subnet in the last octet?
A. 8
B. 32
C. 14
D. 16
ANSWER: D

Why document network and broadcast addresses when subnetting?

A. To change Ethernet speed
B. To replace DHCP
C. To disable routing
D. To identify valid host ranges and avoid invalid address assignments
ANSWER: D

What is Cisco IOS in the context of the syllabus?
A. A cable category
B. A Windows desktop theme
C. Operating system software used on Cisco network devices
D. A DNS record type
ANSWER: C

What is the purpose of a router's command-line interface?
A. To configure, monitor, and troubleshoot the router
B. To assign file ACLs to a PC
C. To edit web pages
D. To create user documents only
ANSWER: A

Which subnet mask supports at least 12 usable IPv4 host addresses?
A. 255.255.255.248 (/29)
B. 255.255.255.0 (/24)
C. 255.255.255.252 (/30)
D. 255.255.255.240 (/28)
ANSWER: D

How many usable host addresses are normally available in a /28 IPv4 subnet?
A. 30
B. 14
C. 16
D. 12
ANSWER: B

What is Cisco Security Device Manager (SDM) generally associated with?
A. A graphical tool for configuring and managing supported Cisco routers
B. A Windows backup format
C. A Linux package manager
D. A wireless encryption algorithm
ANSWER: A

Which mode typically allows configuration changes in Cisco IOS?
A. User EXEC mode only
B. ROM monitor only
C. Global configuration mode or a relevant subconfiguration mode
D. Ping output mode
ANSWER: C

Why should router configuration be saved to startup configuration?
A. To preserve intended settings across a reload
B. To change the MAC address automatically
C. To disable routing protocols
D. To erase the running configuration
ANSWER: A

What should be checked when a subnetting result is validated?
A. Only the switch model
B. Network address, broadcast address, usable range, and prefix
C. Only the router hostname
D. Only cable color
ANSWER: B

For a /28 subnet, how many total addresses are in the block?
A. 14
B. 8
C. 16
D. 32
ANSWER: C

Why divide 152.152.0.0/16 into subnets sized for about 500 hosts each?
A. To make every subnet a /30
B. To remove the need for routing
C. To convert IPv4 to MAC addressing
D. To allocate suitable address blocks while limiting address waste
ANSWER: D

What does a router's boot process do?
A. Assigns every host a DHCP lease
B. Encrypts all routed traffic
C. Loads the operating system and initializes the device
D. Creates Ethernet cables
ANSWER: C

What is the purpose of a configuration register on a router?
A. It influences boot behavior and configuration handling
B. It sets the cable category
C. It stores all Ethernet frames permanently
D. It is a DNS zone identifier
ANSWER: A

Which command is commonly used to test reachability with ICMP echo requests?
A. copy running-config startup-config
B. ping
C. show version only
D. configure terminal
ANSWER: B

Which command-line tool can show the path packets take toward a destination?
A. mkdir
B. passwd
C. format
D. traceroute
ANSWER: D

What is a major security weakness of Telnet?
A. It only works with IPv6
B. It replaces routing protocols
C. It sends session data, including credentials, without encryption by default
D. It always uses strong end-to-end encryption
ANSWER: C

What is hostname resolution?
A. Mapping a host name to an address using mechanisms such as DNS
B. Assigning VLAN IDs automatically
C. Converting a MAC address into a cable type
D. Changing a router's boot image
ANSWER: A

Why use router debugging commands carefully?
A. Debug commands replace backups
B. Debug output cannot affect device performance
C. Debugging always disables routing
D. Detailed debug output can consume resources and produce large volumes of messages
ANSWER: D

Which command commonly displays version and hardware/software details on Cisco IOS?
A. show version
B. show wallpaper
C. show password plaintext
D. show cable category
ANSWER: A

What should be checked if a router cannot resolve a host name?
A. Only the router's screen resolution
B. Only the switch fan speed
C. Only the console cable color
D. Name-server configuration, DNS reachability, and the requested name
ANSWER: D

Why back up router configuration before major changes?
A. To enable recovery if the change causes an outage
B. To replace the router IOS
C. To change all IP addresses automatically
D. To disable the startup configuration
ANSWER: A

What is a static route?
A. A DHCP lease
B. A route manually configured by an administrator
C. A route learned automatically from a routing protocol
D. A switch MAC table entry
ANSWER: B

What is dynamic routing?
A. Routers forward without routing tables
B. Routers learn and update routes through routing protocols
C. Every route is entered manually
D. A method for encrypting Ethernet frames
ANSWER: B

What is the purpose of a routing table?
A. It helps determine the next hop or outgoing interface for IP packets
B. It defines file permissions
C. It lists switch port-security violations only
D. It stores user passwords
ANSWER: A

What is a default route used for?
A. Disabling inter-network communication
B. Replacing all connected routes
C. Forwarding only broadcast frames
D. Forwarding traffic when no more specific route matches
ANSWER: D

Which routing protocol is a link-state protocol commonly taught with OSPF?
A. DHCP
B. OSPF
C. RIP
D. ARP
ANSWER: B

Which routing protocol uses hop count as its classic metric?
A. EIGRP in all configurations
B. RIP
C. Ethernet
D. OSPF
ANSWER: B

What is a routing metric?
A. A cable length label only
B. A user's privilege level
C. A value used to compare possible routes
D. A DNS TTL only
ANSWER: C

What is convergence in dynamic routing?
A. The process by which routers reach a consistent view of routes after a change
B. An IP address becomes a MAC address
C. A switch changes to a new VLAN name
D. A router shuts down permanently
ANSWER: A

What is a next hop in a route?
A. A physical cable category
B. The final application process
C. A DNS alias
D. The next router or forwarding destination toward the network

ANSWER: D

Why verify both routing tables and end-to-end connectivity?
A. A routing table guarantees every application works
B. A route can appear configured while traffic still fails due to other issues
C. Routing tables are used only for wireless
D. Connectivity tests are unrelated to routing
ANSWER: B

What is the primary function of RIP?
A. Assigning IP addresses to clients
B. Encrypting router configuration files
C. Switching Ethernet frames only
D. Exchanging routes using a distance-vector approach
ANSWER: D

What is the classic maximum usable hop count for RIP?
A. 16
B. 15
C. 255
D. 1024
ANSWER: B

What does a RIP metric primarily count?
A. Delay only
B. Bandwidth only
C. Hops
D. VLAN IDs
ANSWER: C

What type of routing protocol is OSPF?
A. Link-state
B. Layer 2 switching protocol
C. Distance-vector only
D. Application-layer
ANSWER: A

What does EIGRP use to select routes?
A. Only the number of VLANs
B. Only DNS response time
C. Only MAC address length
D. A composite metric based on configured factors such as bandwidth and delay
ANSWER: D

What does IGRP stand for?
A. Integrated Group Routing Port
B. Interior Gateway Routing Protocol
C. Internal Graph Resolution Protocol
D. Internet Gateway Relay Process
ANSWER: B

Why configure a static route in a small or controlled topology?
A. It replaces all routing tables
B. It provides explicit, predictable forwarding without route-learning updates
C. It automatically discovers every network
D. It requires no maintenance in any topology
ANSWER: B

What is a neighbor relationship important for in OSPF?
A. It enables routers to exchange routing information and form adjacencies
B. It assigns Ethernet cable categories
C. It performs DHCP address leasing
D. It configures user passwords
ANSWER: A

What should be verified after enabling a routing protocol?
A. Only the console font
B. Only the cable color
C. Only the router hostname

D. Neighbor state where applicable, learned routes, and end-to-end reachability
ANSWER: D

Why can incorrect network statements or interface settings prevent routing updates?
A. It automatically fixes every route
B. The intended interfaces may not participate or may not exchange protocol traffic
C. Routing protocols ignore interfaces
D. Network statements only change wallpaper
ANSWER: B

Which command family on Cisco IOS is commonly used to inspect the IP routing table?
A. show vlan name
B. show user wallpaper
C. show ip route
D. show cable category
ANSWER: C

What does the OSPF cost metric commonly relate to?
A. DNS TTL
B. Interface cost, often derived from bandwidth
C. MAC address vendor
D. Number of users logged in
ANSWER: B

What does an OSPF area help achieve in larger routing designs?
A. Hierarchical organization and reduced routing overhead
B. Eliminating all IP addresses
C. Replacing Ethernet switching
D. Encrypting packets
ANSWER: A

Which protocol is an example of an exterior gateway protocol used between autonomous systems?
A. ARP
B. STP
C. BGP
D. RIP
ANSWER: C

What is the administrative purpose of a static route's next-hop address?
A. It identifies where to forward matching traffic next
B. It sets the DNS search suffix
C. It defines the source MAC vendor
D. It determines cable category
ANSWER: A

What is route redistribution?
A. Encrypting routing tables
B. Exchanging routes between different routing protocols or routing sources
C. Changing a router hostname
D. Converting a VLAN into a cable
ANSWER: B

Why may two routers fail to form a routing-protocol neighbor relationship?
A. Their console cables are different colors
B. A PC has a different wallpaper
C. Protocol settings, network reachability, timers, or interface configuration may mismatch
D. The monitor resolution differs
ANSWER: C

What is a routing loop?
A. A cable twisted pair
B. A condition where packets circulate repeatedly among routers
C. A switch's MAC table aging out normally
D. A DNS record refresh
ANSWER: B

What is the role of a routing protocol's metric?
A. To identify the device manufacturer
B. To compare candidate paths according to protocol rules

C. To set a password policy
D. To allocate a VLAN number
ANSWER: B

After implementing OSPF or EIGRP, what is a useful verification step?
A. Check neighbors, learned routes, and end-to-end traffic
B. Only clear the screen
C. Only inspect cable color
D. Only rename the router
ANSWER: A

What does a Layer 2 switch primarily use to forward Ethernet frames?
A. DNS cache
B. IP routing metric only
C. TCP sequence number
D. MAC address table
ANSWER: D

Why is Spanning Tree Protocol used in switched Ethernet networks?
A. To encrypt Ethernet frames
B. To replace VLANs
C. To assign IP addresses
D. To prevent Layer 2 loops by blocking redundant paths when needed
ANSWER: D

What is the root bridge in STP?
A. The reference switch selected for the spanning-tree topology
B. The router that provides DHCP
C. The switch with the most users always
D. The server that stores DNS zones
ANSWER: A

How does STP bridge ID selection relate to priority?
A. Priority contributes to the bridge ID used in root election
B. Only interface speed determines the root
C. Priority is ignored by STP
D. The highest priority value always wins
ANSWER: A

In classic STP root election, which bridge is preferred when comparing bridge IDs?
A. The bridge with the highest MAC address only
B. The bridge with the lowest bridge ID
C. The bridge with the most ports
D. The bridge with the highest IP address
ANSWER: B

What can happen if STP is disabled in a redundant Layer 2 topology?
A. All routes become static
B. DNS stops resolving every name
C. Every port automatically becomes routed
D. A switching loop and broadcast storm may occur
ANSWER: D

Which is a rapid-convergence STP variant?
A. NAT
B. RIP
C. Rapid Spanning Tree Protocol (RSTP)
D. FTP
ANSWER: C

What is a common reason to change a switch's STP priority?
A. To influence which switch becomes the root bridge
B. To change a PC's IP address
C. To set DNS TTL
D. To increase cable category
ANSWER: A

What is a switching loop likely to cause?
A. Guaranteed lower latency

B. Automatic IP subnetting
C. Encrypted traffic by default
D. Repeated frame circulation and excessive broadcast traffic
ANSWER: D

What should be done before disabling STP on a switch?
A. Clear all IP addresses
B. Assess loop risk and ensure a safe topology/design
C. Remove all VLANs
D. Disable it on every switch without review
ANSWER: B

What is a VLAN?
A. A routing protocol
B. A physical cable type
C. A logical Layer 2 broadcast domain
D. An IPv6 address
ANSWER: C

Why configure VLANs on switches?
A. To encrypt all traffic automatically
B. To segment Layer 2 networks and organize broadcast domains
C. To eliminate the need for IP addresses
D. To replace all routers in every network
ANSWER: B

What is inter-VLAN routing used for?
A. Disabling broadcast domains
B. Creating DHCP leases without a server
C. Forwarding frames only within one VLAN
D. Allowing devices in different VLANs to communicate at Layer 3
ANSWER: D

What is a trunk port commonly used for?
A. Carrying traffic for multiple VLANs between network devices
B. Carrying only one untagged VLAN in every configuration
C. Connecting only serial cables
D. Assigning IP addresses directly
ANSWER: A

What does an access port usually connect to?
A. A DNS root server only
B. A trunk carrying every VLAN by default
C. An end device in a single assigned VLAN
D. A WAN serial circuit only
ANSWER: C

What is VTP used for in Cisco switching environments?
A. Managing router boot images
B. Distributing VLAN configuration information among participating switches
C. Encrypting IPv6 headers
D. Resolving host names
ANSWER: B

What is the usual purpose of a VTP server mode switch?
A. To originate or manage VLAN information in the VTP domain
B. To route all IP packets automatically
C. To disable VLAN propagation
D. To provide DNS for all clients
ANSWER: A

Why must trunking and VLAN membership be checked when same-VLAN PCs cannot ping?
A. Cable category changes VLAN IDs
B. VLANs are unrelated to Layer 2
C. Only DNS can affect ping by IP address
D. A VLAN or trunk mismatch can prevent Layer 2 connectivity
ANSWER: D

What is an extended-range VLAN in common Cisco terminology?

A. A VLAN that uses IPv6 only
B. A VLAN with more than 100 hosts only
C. A VLAN without a switch
D. A VLAN ID from the extended range, typically 1006â€“4094
ANSWER: D

What should be verified after inter-VLAN routing configuration?
A. Only cable color
B. VLAN assignment, trunking, Layer 3 interfaces, gateways, and reachability
C. Only console font
D. Only switch hostname spelling
ANSWER: B

What is the main purpose of an access control list (ACL) on a router?
A. Permit or deny traffic based on configured matching criteria
B. Assign VLAN names automatically
C. Create backups
D. Resolve DNS names
ANSWER: A

What is port security on a switch designed to do?
A. Limit or control which MAC addresses can use a switch port
B. Encrypt every Ethernet frame
C. Replace spanning tree
D. Change the subnet mask of every host
ANSWER: A

What does AAA stand for in network device security?
A. Addressing, Access, and Aggregation
B. Access, Addressing, and Auditing only
C. Authentication, Authorization, and Accounting
D. Authentication, Availability, and ARP
ANSWER: C

What is authentication?
A. Recording usage details only
B. Verifying an identity
C. Deciding what an identity may do
D. Selecting a routing metric
ANSWER: B

What is authorization?
A. Assigning an IP address
B. Recording all logs only
C. Verifying identity only
D. Determining which actions or resources an authenticated identity may access
ANSWER: D

What is accounting in AAA?
A. Recording activity or resource-use information
B. Selecting a root bridge
C. Encrypting all passwords automatically
D. Creating VLAN trunks
ANSWER: A

Which protocol commonly uses TCP port 49 for centralized device administration?
A. DNS
B. RADIUS
C. TACACS+
D. DHCP
ANSWER: C

Which transport protocol is traditionally used by RADIUS?
A. ARP
B. ICMP
C. GRE only
D. UDP
ANSWER: D

Why use infrastructure device hardening?
A. To disable logging
B. To allow all management access publicly
C. To share administrator credentials
D. To reduce unnecessary services, access, and attack surface
ANSWER: D

What is a good ACL verification practice?
A. Test permitted and denied traffic and inspect rule order/counters
B. Assume the ACL works from its name
C. Remove the implicit deny concept from consideration
D. Disable all routing
ANSWER: A

What is an implicit deny in a typical ACL?
A. It creates a VLAN
B. All unmatched traffic is automatically permitted
C. Traffic not matched by an earlier permit rule is denied
D. It disables the router
ANSWER: C

Why does ACL rule order matter?
A. Order affects only the hostname
B. ACLs ignore rule order
C. Rules are sorted by cable length
D. Rules are evaluated in order and an earlier match can determine the result
ANSWER: D

What is an access-layer threat mitigation example?
A. Disabling all switch logs
B. Port security and controls against unauthorized network access
C. Removing every VLAN
D. Sharing admin passwords
ANSWER: B

What is a MAC address sticky feature used for in supported switch port security?
A. Assigning an IPv6 prefix
B. Creating a GRE tunnel
C. Resolving DNS names
D. Learning and associating observed MAC addresses with the port configuration
ANSWER: D

Which AAA protocol commonly combines authentication and authorization in a UDP-based exchange?
A. RADIUS
B. OSPF
C. STP
D. TACACS+ only over TCP port 49
ANSWER: A

What is a commonly cited TACACS+ characteristic?
A. It is a wireless cable standard
B. It uses TCP and separates authentication, authorization, and accounting functions
C. It assigns DHCP addresses
D. It is a Layer 2 loop-prevention protocol
ANSWER: B

Why should management access use secure protocols rather than Telnet where possible?
A. Telnet encrypts all data by default
B. Secure protocols protect credentials and session data in transit
C. It makes ACLs unnecessary
D. Secure protocols disable authentication
ANSWER: B

What is a security benefit of disabling unused switch ports?
A. It replaces port security
B. It reduces opportunities for unauthorized physical network access
C. It increases the subnet size
D. It enables every VLAN automatically
ANSWER: B

What should be checked when an ACL unexpectedly blocks traffic?
A. Only the switch model year
B. Only the device wallpaper
C. Source/destination, protocol, port, direction, and rule order
D. Only the cable label
ANSWER: C

Why centralize AAA where appropriate?
A. It removes the need for identity checks
B. It grants all users admin rights
C. It can improve consistent access control and auditing across devices
D. It prevents all network outages
ANSWER: C

What is Network Address Translation (NAT) used for?
A. Replacing routing tables
B. Converting TCP to UDP
C. Assigning VLAN names
D. Translating IP address information between addressing realms
ANSWER: D

What is PAT commonly called?
A. Static IPv6
B. Dynamic DNS
C. NAT overload
D. Spanning tree
ANSWER: C

Why is NAT commonly used in IPv4 networks?
A. It eliminates routing
B. It replaces all firewalls
C. It provides end-to-end encryption
D. It can allow multiple private hosts to share public address resources
ANSWER: D

What is static NAT?
A. A routing protocol
B. A fixed mapping between an inside and outside address
C. An IPv6 multicast address
D. A dynamic mapping chosen from a pool only
ANSWER: B

How many bits are in an IPv6 address?
A. 48 bits
B. 128 bits
C. 64 bits
D. 32 bits
ANSWER: B

Which notation is commonly used to represent IPv6 addresses?
A. MAC notation only
B. Dotted decimal only
C. Colon-separated hexadecimal groups
D. Base-10 octets only
ANSWER: C

What does the IPv6 loopback address look like?
A. 192.168.1.1
B. 0.0.0.0
C. ::1
D. 255.255.255.255
ANSWER: C

What is one major reason IPv6 was developed?
A. To remove packet headers
B. To eliminate all routing protocols
C. To replace Ethernet cabling
D. To provide a much larger address space than IPv4

ANSWER: D

What is a common IPv6 link-local prefix?
A. 192.168.0.0/16
B. 127.0.0.0/8
C. FE80::/10
D. 200.0.0.0/8
ANSWER: C

What should be checked when configuring IPv6 on a router?
A. Only cable category
B. Only IPv4 subnet mask
C. Only the switch root priority
D. IPv6 addressing, interface state, routing, and reachability
ANSWER: D

Which NAT type typically maps one private address to one public address persistently?
A. Dynamic routing
B. PAT only
C. NAT-free routing
D. Static NAT
ANSWER: D

What information does PAT use to distinguish many translated sessions sharing an address?
A. Ethernet cable categories
B. DNS TTL values
C. Transport-layer port numbers
D. VLAN names only
ANSWER: C

What is an inside local address in common NAT terminology?
A. An IPv6 multicast address
B. The address assigned to an inside host as seen on the inside network
C. The public destination address only
D. The router's MAC address
ANSWER: B

Which IPv6 address type is intended for one-to-nearest interface delivery within a scope?
A. Broadcast
B. IPv4 loopback
C. Anycast
D. Ethernet multicast only
ANSWER: C

Does IPv6 use broadcast addresses in the same way as IPv4?
A. Only when NAT is enabled
B. Yes, using 255.255.255.255
C. Yes, every IPv6 subnet has an all-ones broadcast
D. No; IPv6 uses multicast for many functions instead of broadcast
ANSWER: D

What does the IPv6 prefix length /64 indicate?
A. The first 64 bits are the network prefix
B. A 64-octet address
C. A 64-bit total IPv6 address
D. 64 host addresses exactly
ANSWER: A

What is SLAAC used for in IPv6?
A. Translating IPv4 ports
B. Creating STP root bridges
C. Preventing all routing
D. Stateless address autoconfiguration by hosts
ANSWER: D

Which protocol is important for IPv6 neighbor discovery?
A. RIP version 1
B. FTP
C. TACACS+

D. ICMPv6
ANSWER: D

Why may NAT not be required for ordinary IPv6 address conservation in the same way as IPv4?
A. IPv6 cannot route packets
B. IPv6 has no addresses
C. IPv6 uses only private addresses
D. IPv6 provides a vastly larger address space
ANSWER: D

What should be verified after applying NAT rules?
A. Only the DNS MX record
B. Only the switch hostname
C. Translation entries, routing, ACL interactions, and application connectivity
D. Only the cable category
ANSWER: C

What is a WAN?
A. A Layer 2 access port only
B. A DNS record
C. A single computer's file system
D. A network that connects sites across a broad geographic area
ANSWER: D

What does PPP provide?
A. A web application framework
B. A routing metric only
C. A data-link protocol for point-to-point links
D. A file permission model
ANSWER: C

What does MLPPP allow?
A. Encrypting DNS records
B. Creating multiple VLAN names only
C. Converting IPv4 to IPv6
D. Combining multiple physical links into one logical PPP bundle
ANSWER: D

What does PPPoE stand for?
A. Packet Path Protocol over Ethernet
B. Point-to-Point Protocol over Ethernet
C. Private Port Protocol over Exchange
D. Point-to-Point Proxy of Encryption
ANSWER: B

What is local authentication in a PPP lab commonly used for?
A. Choosing the STP root bridge
B. Authenticating the peer using locally configured credentials
C. Assigning VLAN tags
D. Resolving DNS names
ANSWER: B

What is a GRE tunnel used for?
A. Replacing the IP routing table
B. Encapsulating one network protocol inside another to create a virtual tunnel
C. Encrypting all traffic by itself
D. Providing DHCP leases
ANSWER: B

Does basic GRE inherently provide encryption?
A. No; GRE itself does not provide encryption
B. Yes, by using DNS
C. Only when STP is enabled
D. Yes, always with AES
ANSWER: A

What is eBGP used for?
A. Resolving local host names
B. Exchanging routing information between different autonomous systems

C. Switching frames within one VLAN
D. Configuring Ethernet cable categories
ANSWER: B

What is a BGP autonomous system number associated with?
A. A VLAN name
B. Identifying an autonomous system in inter-domain routing
C. A TCP port number
D. A MAC address vendor code
ANSWER: B

What should be verified when troubleshooting PPPoE connectivity?
A. Only the STP priority
B. Only the DNS MX record
C. Physical/Ethernet path, PPPoE session, authentication, and IP configuration
D. Only the desktop theme
ANSWER: C

What does SDN stand for?
A. System Device Network
B. Software-Defined Networking
C. Secure Data Naming
D. Switched Domain Numbering
ANSWER: B

What is the control plane responsible for in SDN?
A. Storing user files
B. Making forwarding decisions and managing network behavior
C. Providing the physical cable
D. Forwarding every packet in hardware only
ANSWER: B

What is the data plane responsible for?
A. Writing all network policies by itself
B. Forwarding packets according to installed forwarding rules
C. Installing operating systems
D. Managing user passwords
ANSWER: B

What is a common SDN architectural characteristic?
A. Removing the need for forwarding devices
B. Replacing IP with file paths
C. Separating control logic from packet-forwarding functions
D. Combining every network function into a single cable
ANSWER: C

Why is scalability important in data-centre or ISP networks?
A. The network must handle growth in devices, traffic, and policy complexity
B. It removes the need for monitoring
C. It determines the cable color
D. It makes addressing unnecessary
ANSWER: A

What does network reliability concern?
A. Consistent service operation and availability despite failures
B. Only the VLAN naming style
C. Only the administrator's password length
D. Only the number of switch LEDs
ANSWER: A

What does QoS help manage?
A. DNS zone transfers only
B. Operating system installation
C. Traffic treatment such as priority, delay, jitter, or bandwidth
D. User file permissions
ANSWER: C

What does configuration consistency mean in network management?
A. Every device must use the same IP address

B. All traffic must use one protocol
C. Devices and policies remain aligned with intended configurations
D. Logs must be disabled
ANSWER: C

How can access control violations affect SDN consistency?
A. They automatically improve policy correctness
B. They only affect cable shielding
C. Unauthorized or conflicting changes can cause policy deviations
D. They cannot influence network behavior
ANSWER: C

Which is a recognized challenge in SDN adoption?
A. It requires no planning
B. Controller resilience, security, interoperability, and operational complexity
C. It removes every possible network fault
D. It eliminates the need for skills
ANSWER: B

What is virtual networking?
A. Creating logical network connectivity using software-defined components
B. A database indexing method
C. A file compression method
D. Replacing all physical infrastructure with cables
ANSWER: A

What is a virtual switch?
A. A physical-only router interface
B. Software that forwards traffic among virtual machines and/or physical networks
C. A DNS zone
D. A disk partition
ANSWER: B

What is one use case for network access control (NAC)?
A. Formatting virtual disks
B. Applying access policies based on device or user compliance and identity
C. Choosing cable category
D. Changing a file extension
ANSWER: B

What is a virtual customer edge (vCPE) conceptually?
A. Network edge functions delivered using virtualized software at or near the customer edge
B. A local user account
C. A physical cable tester
D. A DNS reverse zone
ANSWER: A

How can virtual networking help data-centre optimization?
A. It guarantees no packet loss
B. It removes all security controls
C. It can make network provisioning and segmentation more flexible
D. It eliminates the need for IP addressing
ANSWER: C

What is a benefit of separating logical networks?
A. It removes all routing requirements
B. Every workload must share one broadcast domain
C. Different workloads can be segmented through software-defined policies
D. It guarantees encryption
ANSWER: C

What is a common concern when configuring virtual switches?
A. Printer paper size
B. Keyboard language
C. Correct port groups, VLAN tagging, uplinks, and isolation
D. Desktop wallpaper resolution
ANSWER: C

Why should virtual network changes be tested?

A. It automatically backs up all VMs
B. Virtual networks cannot affect workloads
C. Testing changes only the VM name
D. Misconfiguration can interrupt connectivity for multiple virtual workloads
ANSWER: D

What does a virtual switch uplink commonly connect to?
A. A physical network interface or upstream network
B. A CPU register
C. A DNS TXT record only
D. A user password file
ANSWER: A

What should be validated after deploying a virtual network?
A. Connectivity, segmentation, VLAN handling, and security policy
B. Only the host name spelling
C. Only the VM wallpaper
D. Only the terminal color
ANSWER: A

What is Mininet commonly used for?
A. Emulating virtual hosts, switches, links, and SDN network topologies
B. Creating email mailboxes
C. Formatting physical disks
D. Managing Windows domain users
ANSWER: A

What is a linear topology in a network simulation?
A. Every node connected to every other node
B. A hierarchy with a root and branches only
C. Nodes connected in a chain-like sequence
D. A topology with no links
ANSWER: C

What is a tree topology?
A. A hierarchical structure with branches from parent nodes
B. A single unbranched line only
C. A topology without switches
D. A full mesh by definition
ANSWER: A

Why run ping tests in Mininet?
A. To install Mininet packages
B. To configure cable shielding
C. To create a new user account
D. To check basic IP reachability between simulated hosts
ANSWER: D

What is a Mininet host?
A. A simulated end system that can run network applications or commands
B. A physical router only
C. A DNS record
D. A switch priority value
ANSWER: A

What is a Mininet switch?
A. A cable category
B. A user identity
C. A file permission
D. A simulated switch element that forwards traffic in the emulated topology
ANSWER: D

What is a link in a Mininet topology?
A. A connection between network elements
B. A user password
C. A routing metric only
D. A DNS cache entry
ANSWER: A

Why verify installation before creating a topology?
A. To remove the need for a topology
B. To guarantee every ping succeeds
C. To disable the controller
D. To ensure required tools and dependencies are available
ANSWER: D

If ping fails between Mininet hosts, what should be checked?
A. Only the keyboard layout
B. Only the host's terminal color
C. Only the file name
D. Host addresses, topology links, switch/controller behavior, and routing
ANSWER: D

What is the purpose of simulating a network before deployment?
A. To guarantee production will never fail
B. To explore behavior and test configurations in a controlled environment
C. To remove the need for documentation
D. To replace all real-world validation
ANSWER: B